Comparing Models for Glioma Grade Classification

DSC 201 - Project 2 | Tarun Vigneswaran

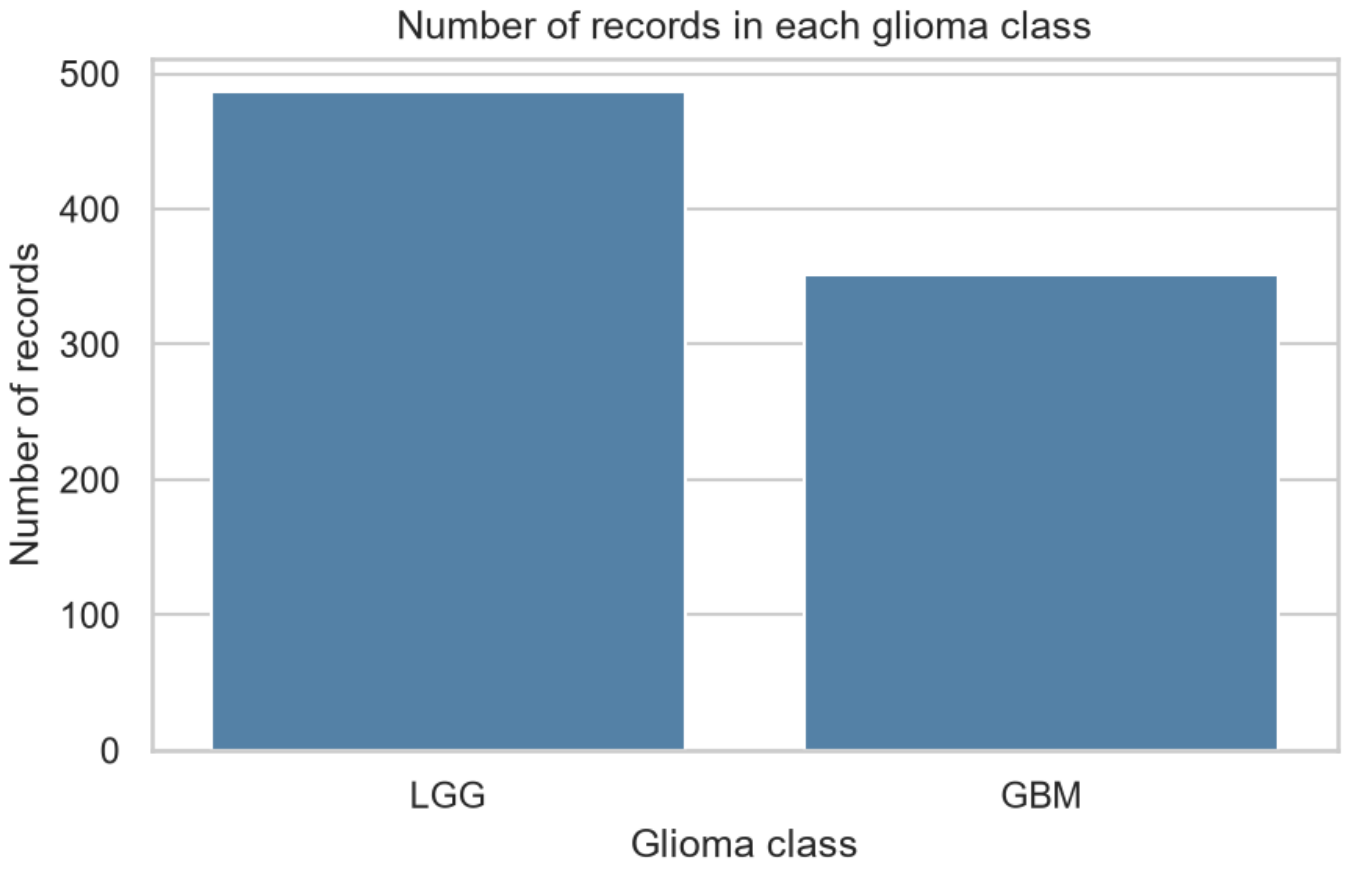
1. Introduction

This project compares Logistic Regression, Random Forest, AdaBoost, and XGBoost using the approved UCI Glioma Grading Clinical and Mutation Features dataset. The original file has 839 records and 23 input features: 20 gene-mutation features and three clinical features (age, gender, and race). The target is Grade: 0 means Lower-Grade Glioma (LGG), and 1 means Glioblastoma (GBM). We predict the dataset's grade labels among glioma records, not whether a person has cancer. This is an educational comparison, not a medical diagnostic system.

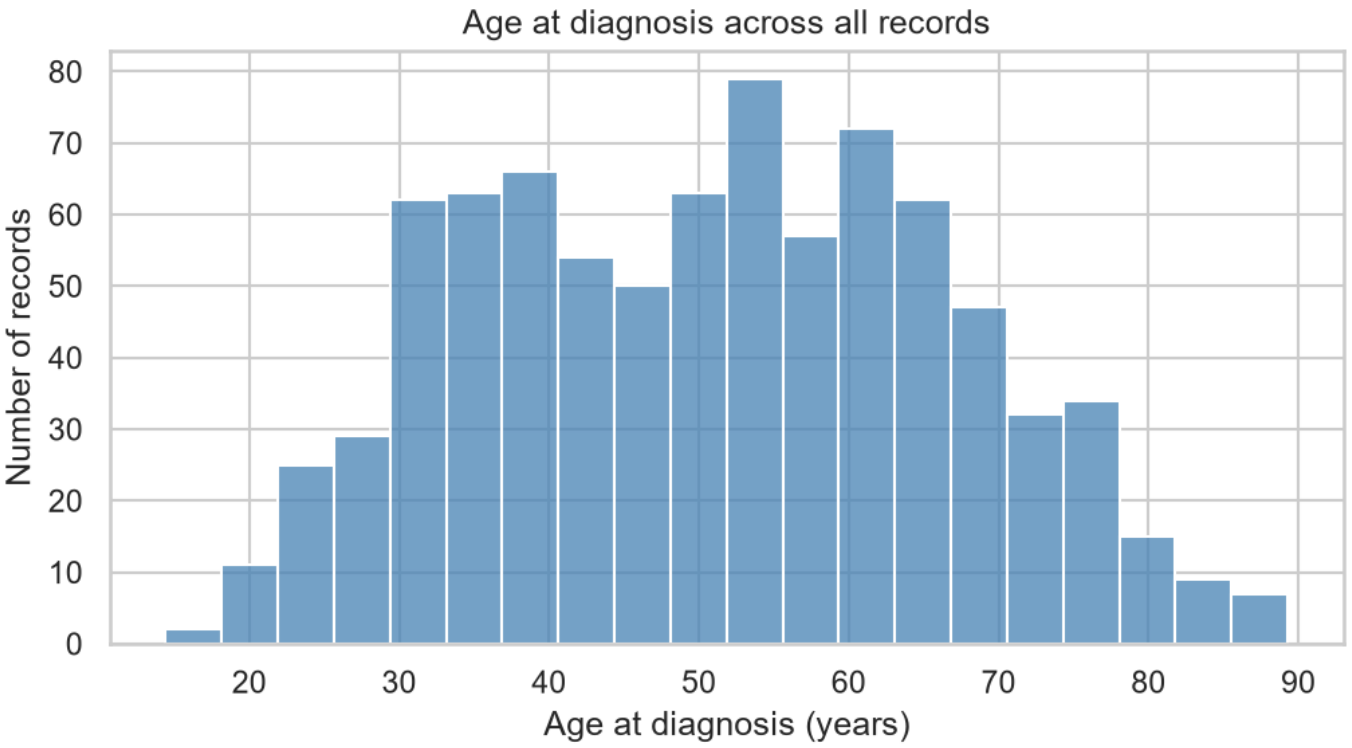
2. Data exploration and preprocessing

The original data contains 487 LGG and 352 GBM records. There are no missing values, so no values were filled in. There is one exact repeated record. I kept the first copy and removed the repeat before splitting, leaving 838 records: 486 LGG and 352 GBM. Without patient IDs, I cannot prove those two records were the same person. Removing the repeat avoids identical records appearing in both training and testing, but it could remove a different person with matching values. The downloaded CSV stays unchanged.

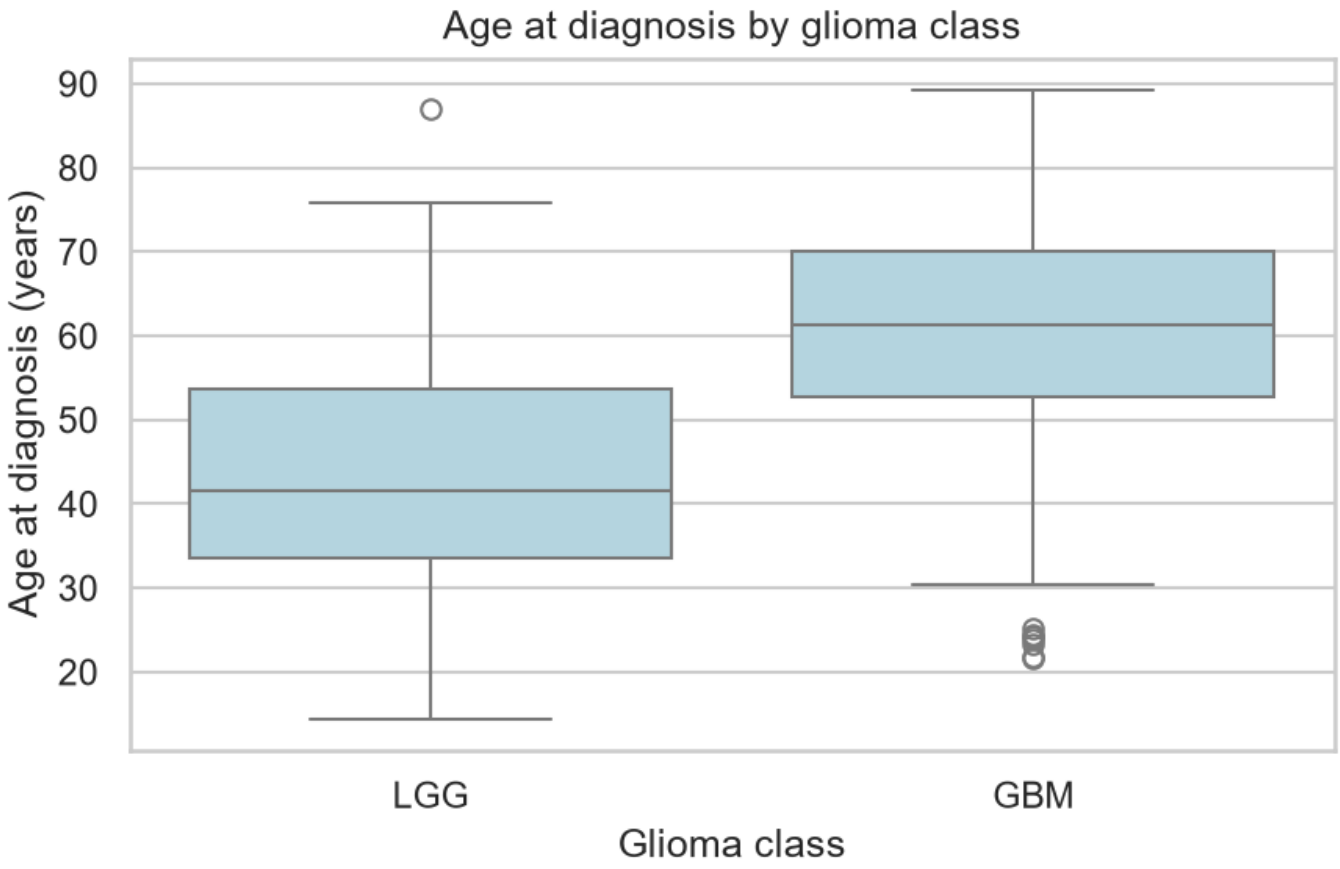
Age ranges from 14.42 to 89.29 years. The 1.5-times-IQR check flags no ages in the prepared data, so none were removed or changed. The original median age is 41.61 for LGG and 61.395 for GBM. GBM cases tend to be older here, but the ranges overlap. Age alone cannot separate the groups perfectly.



Class counts

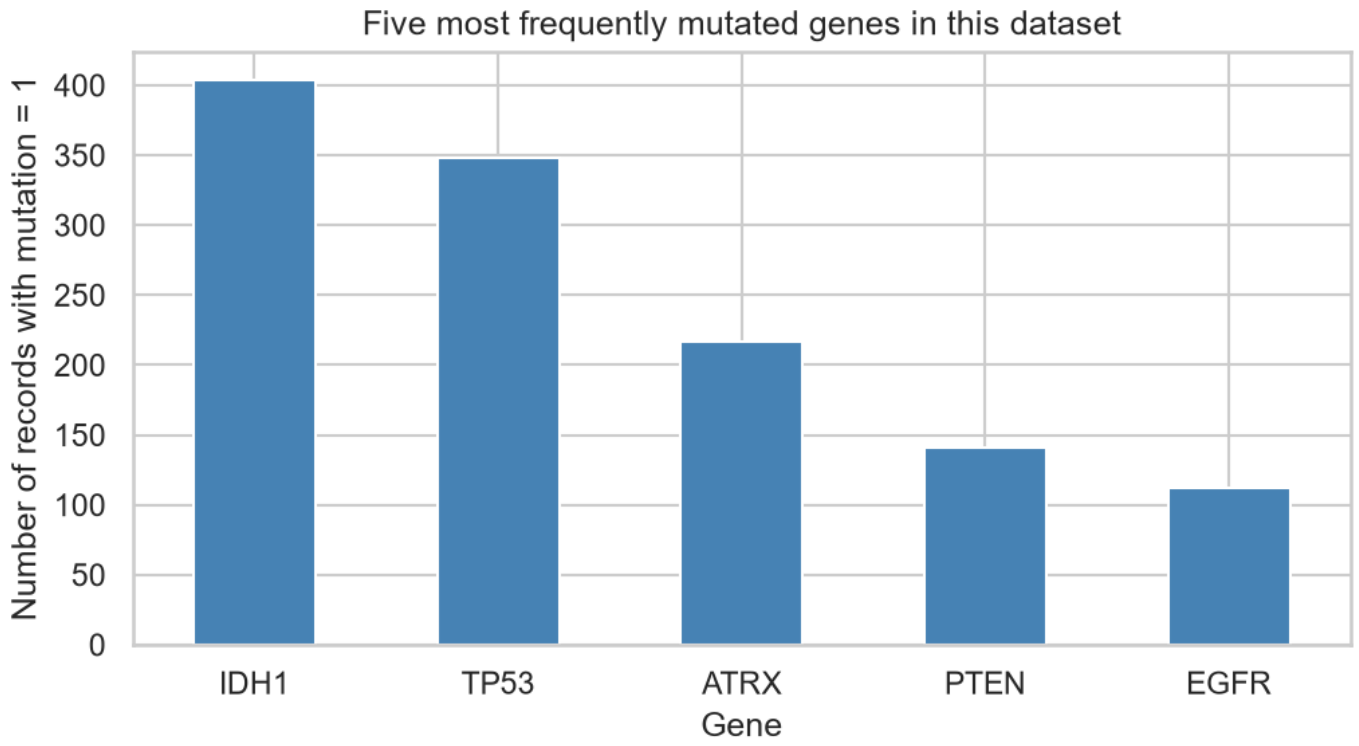


Age distribution

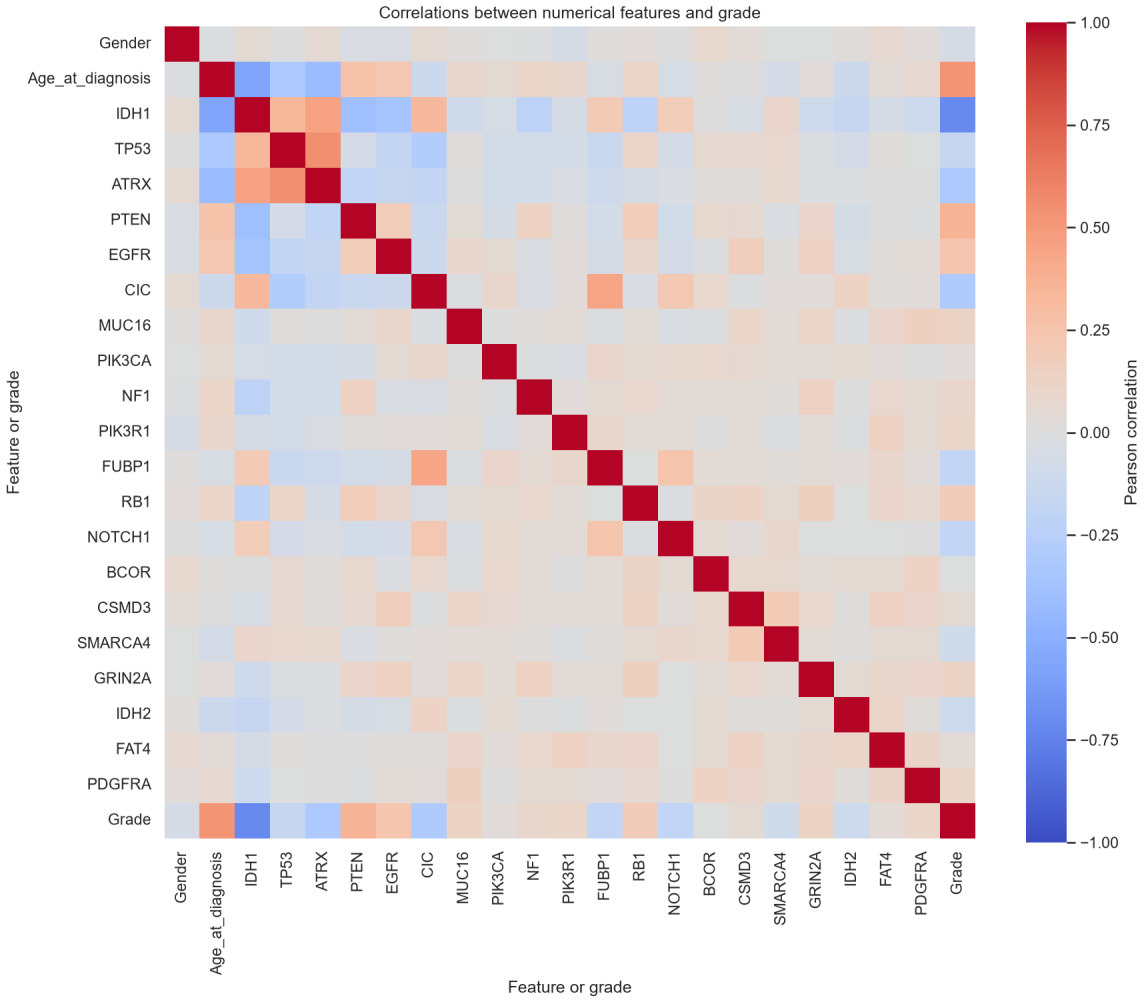


Age by class

The five most frequently mutated genes are IDH1 (404 records), TP53 (348), ATRX (217), PTEN (141), and EGFR (112). Frequency is not the same as predictive importance. All 20 gene features were kept. In the original file, IDH1 and Grade have a correlation of about -0.708, while age and Grade have a correlation of about 0.529. These describe this sample and do not prove cause and effect.



Mutation frequency



Correlations



Age and mutation scatter plot

The scatter plot shows age and the number of mutations among the 20 recorded genes. This count is used only for the plot, not added as a model feature. The two classes overlap in the chart.

Race arrives as four text labels, even though UCI's variable description lists number codes. Each race category becomes a separate yes/no column, avoiding an artificial numerical order. The category list comes from UCI rather than learned test outcomes. There are 26 numeric input columns after encoding the original 23 features. Gender and the genes already use 0 and 1. Grade is never an input.

The reusable split function reserves 20% for testing, keeps similar class proportions, and accepts a reproducible seed. It also supports optional training-only standardization. For model searches, I leave that option off and put scaling inside the Logistic Regression pipeline. Each CV training fold therefore learns its own scaling values. Tree models do not need scaling.

3. Models and tuning

- **Logistic Regression:** learns a weighted rule for GBM. I tuned C (0.1, 1, 10), which controls how strongly large coefficients are limited.

- **Random Forest:** combines many decision trees. I tuned the tree count (50, 100) and maximum depth (5 or unlimited).

- **AdaBoost:** adds small trees and gives more weight to difficult cases. I tuned the tree count (30, 60) and learning rate (0.1, 0.5).

- **XGBoost:** adds trees to improve earlier predictions. I used 50 trees and tuned maximum depth (2, 3) and learning rate (0.05, 0.1).

Each grid has three or four combinations. All models use GridSearchCV with the same five shuffled stratified folds and F1 as the tuning score. GBM (1) is the positive class. Precision asks how many predicted GBM cases were correct; recall asks how many actual GBM cases were found. F1 balances precision and recall. Undefined precision or recall is set to zero using zero_division=0.

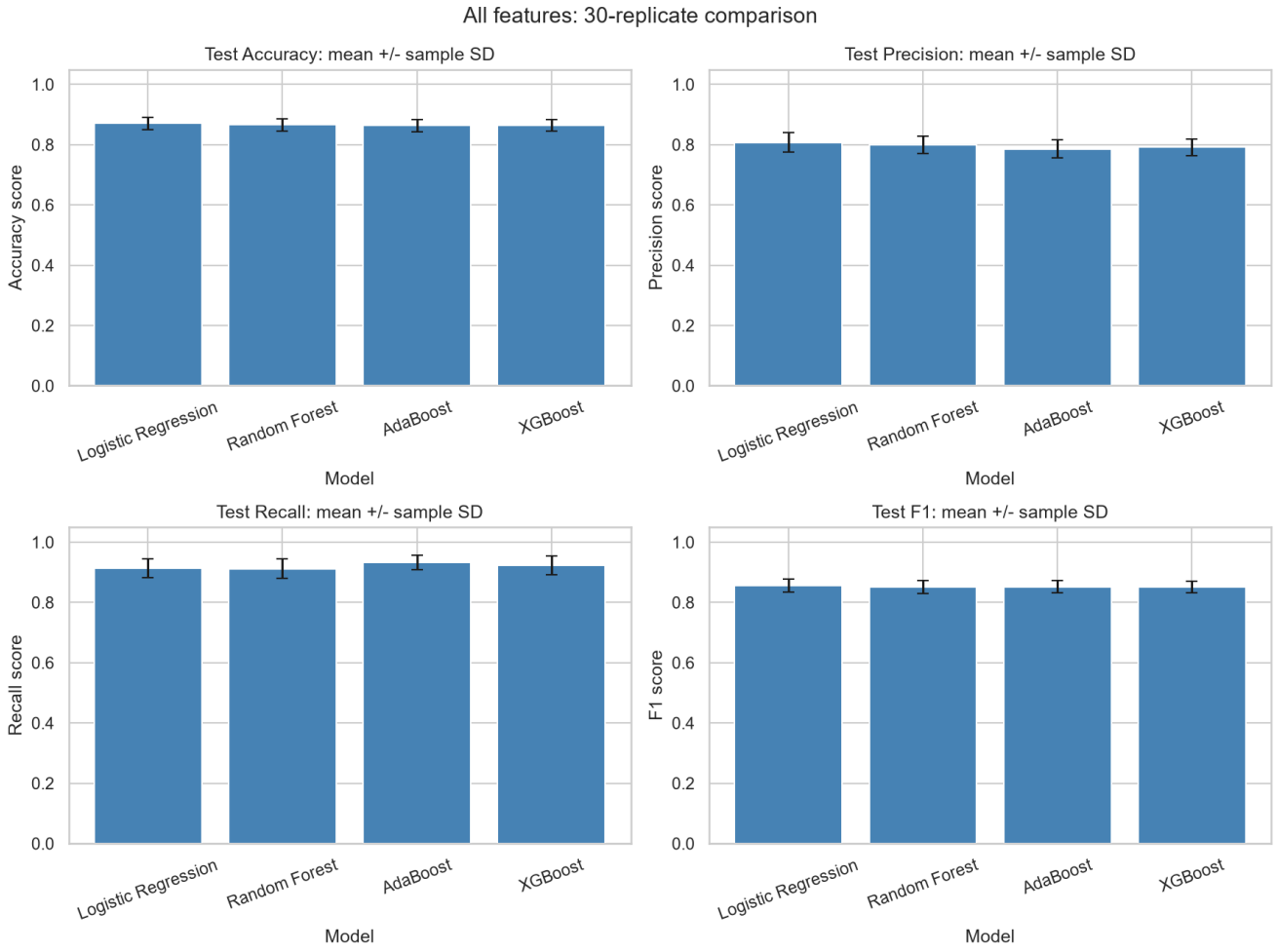
For each seed from 100 to 129, I made a new stratified 80/20 split: 670 training records and 168 test records. All four models used the same people and folds within each repetition. The search used training data only, then automatically refitted its best pipeline on the whole training set. Test scores were calculated after that. The complete search was repeated 30 times, not tuned once and reused.

4. Evaluation and visualization

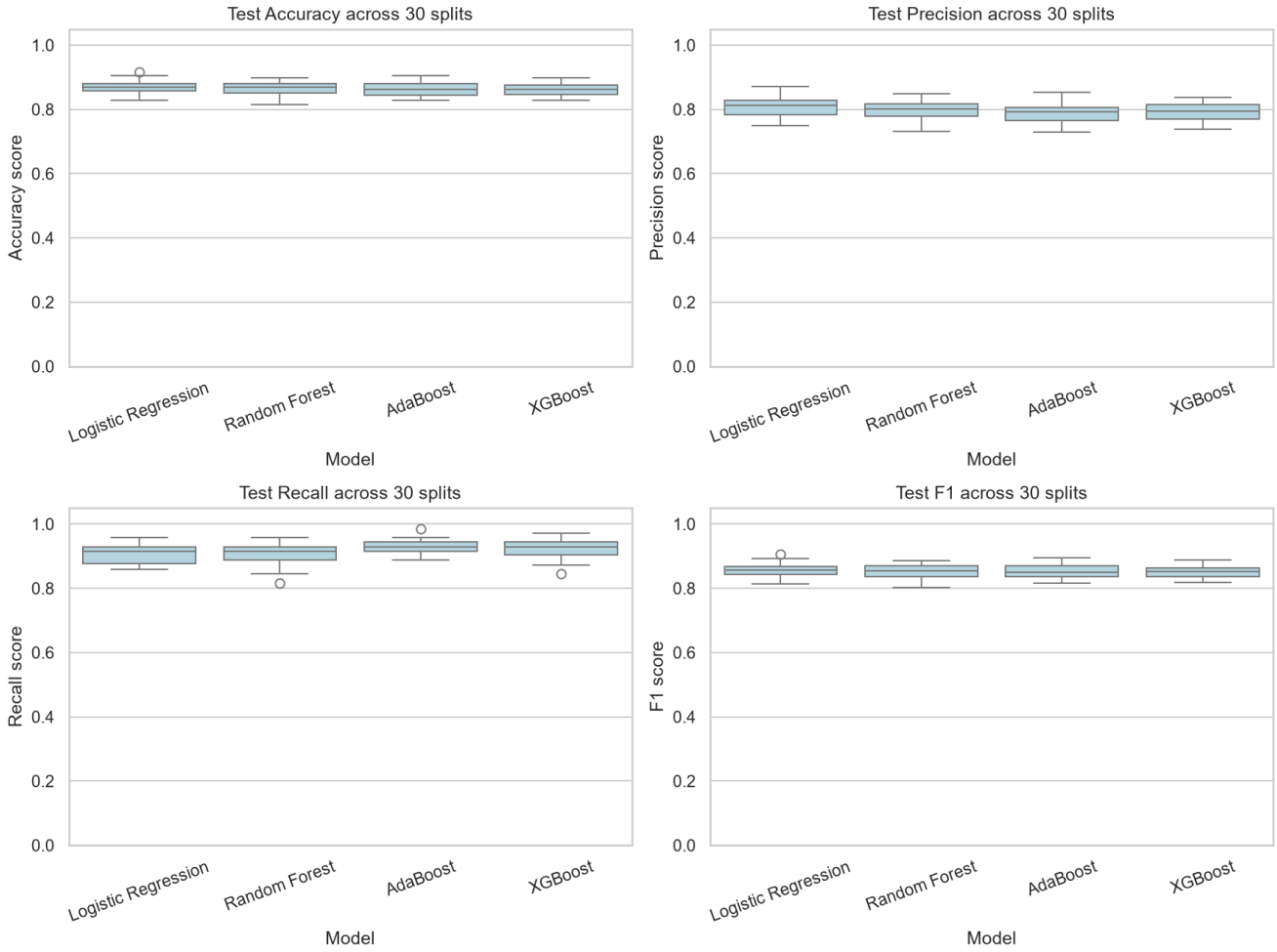
The following are held-out test scores for all features, on a 0-1 scale. The values after +/- are sample standard deviations across the 30 repetitions.

Model	Accuracy (mean +/- SD)	Precision (mean +/- SD)	Recall (mean +/- SD)	F1 (mean +/- SD)
Logistic Regression	0.8706 +/- 0.0211	0.8078 +/- 0.0329	0.9131 +/- 0.0312	0.8566 +/- 0.0219
Random Forest	0.8655 +/- 0.0194	0.7996 +/- 0.0284	0.9117 +/- 0.0327	0.8514 +/- 0.0208
AdaBoost	0.8633 +/- 0.0205	0.7860 +/- 0.0303	0.9319 +/- 0.0240	0.8523 +/- 0.0201
XGBoost	0.8641 +/- 0.0182	0.7916 +/- 0.0273	0.9230 +/- 0.0307	0.8517 +/- 0.0187

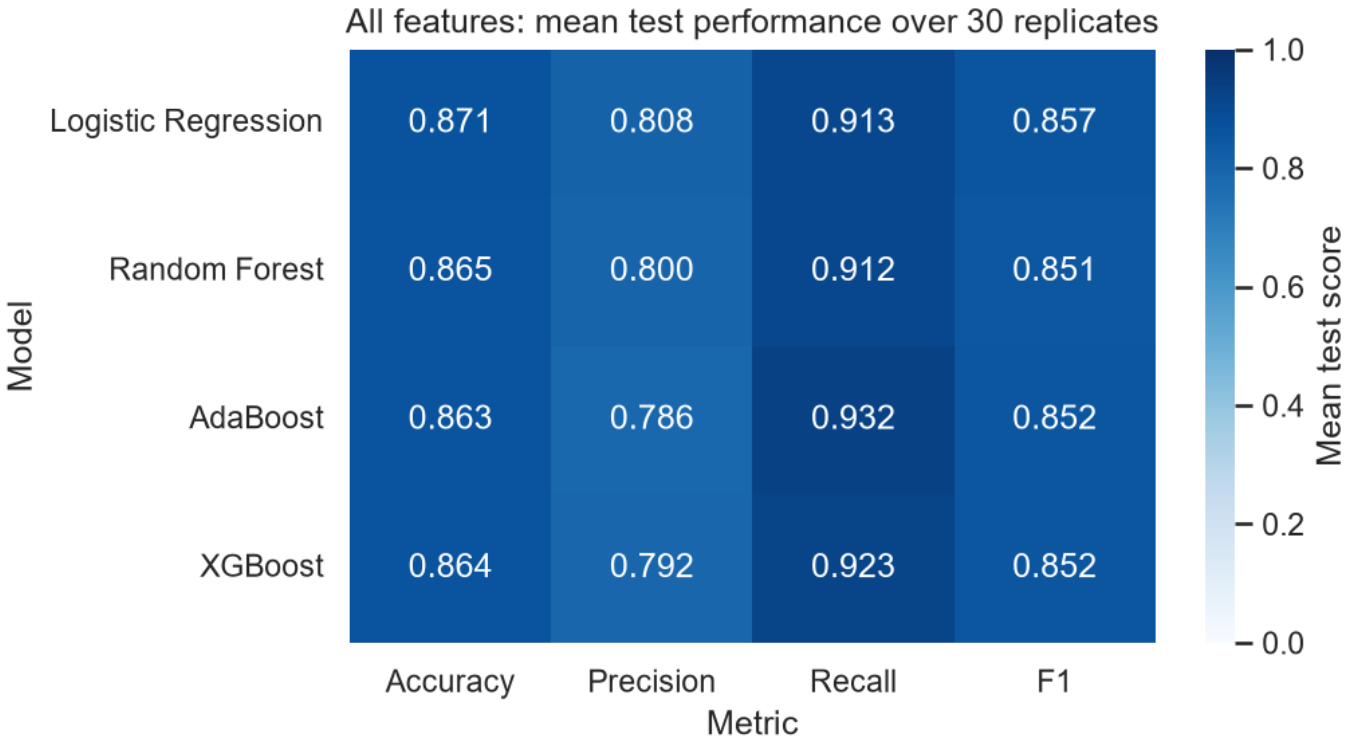
Logistic Regression had the highest mean F1, 0.8566, and mean accuracy 0.8706. For context, always predicting LGG would give 97/168 = 0.5774 accuracy in these test splits, but zero GBM recall. Accuracy alone would not show that failure to identify GBM.



Mean scores and SD



Score distributions



Mean score summary

The bar charts show averages and SD. The box plots show how scores change between splits. The heatmap summarizes the average scores. Minimum and maximum scores show the observed range, not guaranteed limits for future data:

Model	Test metric	Minimum	Maximum
Logistic Regression	Accuracy	0.8274	0.9167
Logistic Regression	Precision	0.7500	0.8701
Logistic Regression	Recall	0.8592	0.9577
Logistic Regression	F1	0.8129	0.9054
Random Forest	Accuracy	0.8155	0.8988
Random Forest	Precision	0.7326	0.8493
Random Forest	Recall	0.8169	0.9577
Random Forest	F1	0.8025	0.8859
AdaBoost	Accuracy	0.8274	0.9048
AdaBoost	Precision	0.7283	0.8533
AdaBoost	Recall	0.8873	0.9859
AdaBoost	F1	0.8153	0.8947
XGBoost	Accuracy	0.8274	0.8988

Model	Test metric	Minimum	Maximum
XGBoost	Precision	0.7386	0.8378
XGBoost	Recall	0.8451	0.9718
XGBoost	F1	0.8176	0.8889

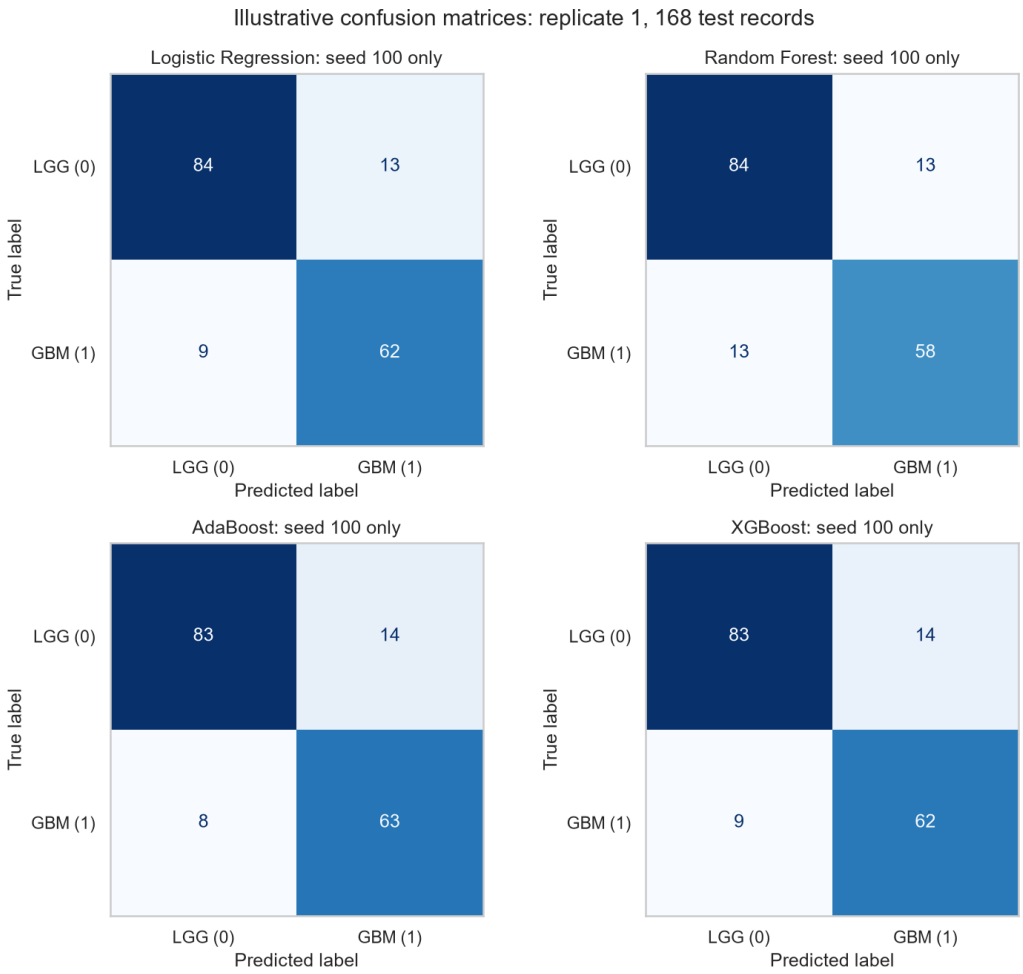
For comparison, the table below keeps training averages separate from test averages:

Model	Training accuracy	Test accuracy	Training F1	Test F1
Logistic Regression	0.8780	0.8706	0.8626	0.8566
Random Forest	0.8819	0.8655	0.8679	0.8514
AdaBoost	0.8693	0.8633	0.8566	0.8523
XGBoost	0.8742	0.8641	0.8610	0.8517

Higher training than test scores can signal that the model has learned details that do not carry over to new records. A small gap also does not prove readiness for practical use.

Illustrative confusion matrices

These matrices show **only replicate 1, seed 100**, selected before seeing its scores. Its test set has 97 LGG and 71 GBM records. They are not pooled matrices for all 30 repetitions. Individual predictions are saved in the results folder.



Seed 100 confusion matrices

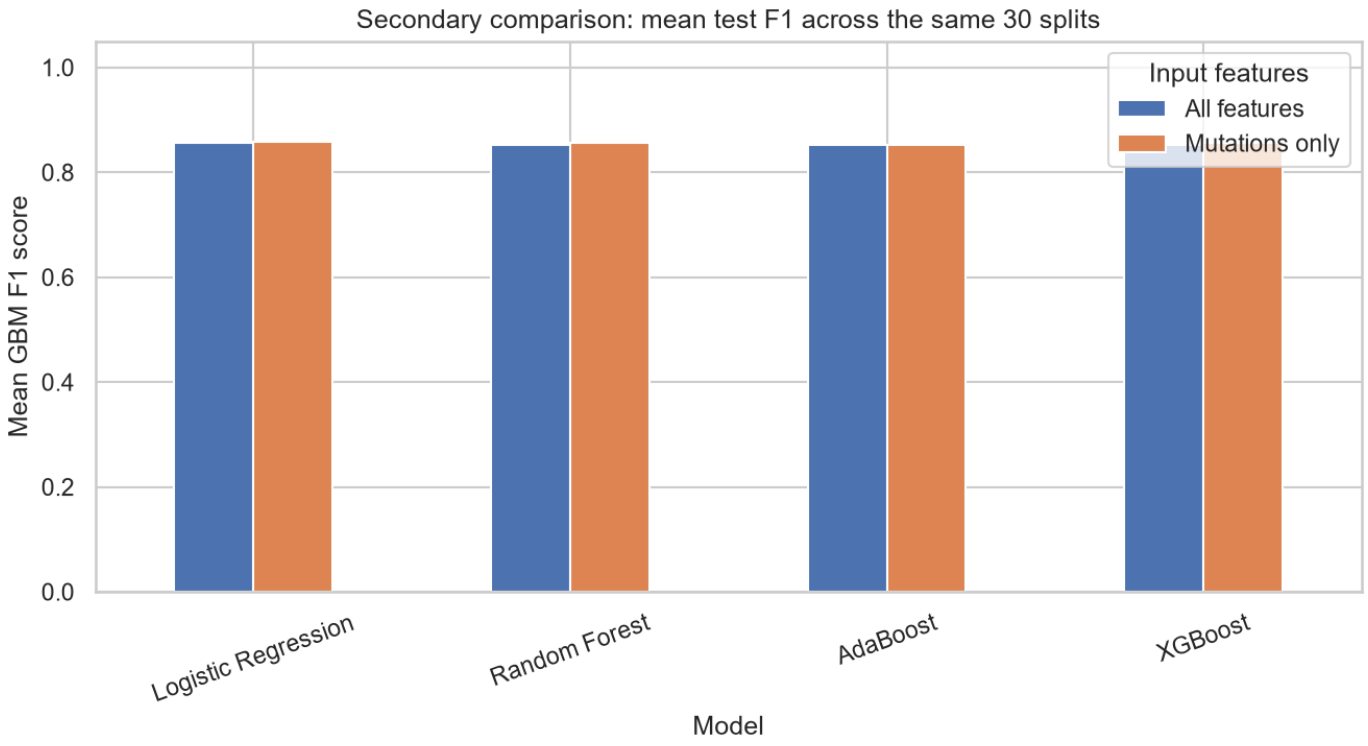
Model	Correct LGG	LGG predicted GBM	GBM predicted LGG	Correct GBM
Logistic Regression	84	13	9	62
Random Forest	84	13	13	58
AdaBoost	83	14	8	63
XGBoost	83	14	9	62

5. Secondary comparison: mutation features only

I repeated the same searches with only the 20 recorded mutation genes, using the same people and splits. No random gene selection was used.

Model	All-feature mean F1	Gene-only mean F1	All minus genes
Logistic Regression	0.8566	0.8578	-0.0012
Random Forest	0.8514	0.8563	-0.0049
AdaBoost	0.8523	0.8517	+0.0006

Model	All-feature mean F1	Gene-only mean F1	All minus genes
XGBoost	0.8517	0.8563	-0.0046



Feature-set comparison

Adding clinical features raised mean F1 for 1 of the four models in these runs. More features therefore did not give a universal improvement. These small differences should be read alongside split-to-split variation, not treated as proof that one feature set is always better.

6. Findings, limitations, and conclusion

By the chosen comparison measure, mean test F1, Logistic Regression ranked first among the four all-feature models. A simple weighted rule may fit the main patterns without learning as much noise as a more flexible model. This is a possible explanation, not a proven reason. Its mean F1 lead over the next model is only 0.0043; this project does not establish a statistically reliable winner. Different grids, data, or seeds could change the ranking.

The sample is small and comes from one published dataset. Class sizes differ, and some race groups have very few records (including one record in one group). Overall scores do not establish fairness across demographic groups. The dataset does not provide enough evidence for reliable subgroup conclusions here. Clinical and genetic records also require privacy protection.

The 30 test groups overlap, so the SD describes changes across splits, not 30 independent studies or a confidence interval. Comparing models using these test results does not replace evaluation on a new external dataset. Shifts in who is sampled, measurement methods, or healthcare settings could change results. No external validation, clinical safety assessment, or fairness study was done.

The project shows how to compare simple tuned classifiers using the same data splits. All four models were actually trained and evaluated. The scores are useful for this classroom comparison, but none should make medical decisions on its own.

Source and reproducibility

Tasci, E., Camphausen, K., Krauze, A. V., and Zhuge, Y. (2022). *Glioma Grading Clinical and Mutation Features*. UCI Machine Learning Repository. Dataset DOI.

Data was loaded through ucimlrepo, dataset ID 759. Model versions are listed in requirements.txt. See README for running instructions, individual results, and summary statistics. The original data is kept in glioma.csv. This draft reflects the saved configuration; review it if you change the code or rerun with different settings.